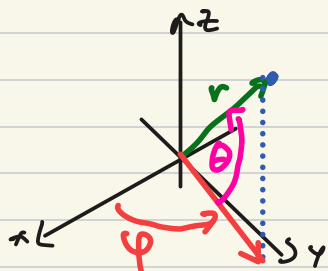


April 15, 2026

Caution!: Last lecture we defined spherical coordinates with the convention: r - radius, θ - elevation angle, φ - azimuthal angle

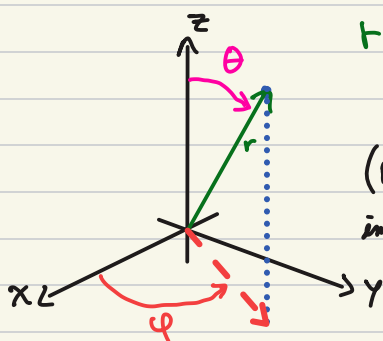


$$(r, \theta, \varphi) \leftrightarrow \begin{pmatrix} r \cos \theta \cos \varphi \\ r \cos \theta \sin \varphi \\ r \sin \theta \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

in polar in cartesian

$$0 \leq r, \quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}, \quad 0 \leq \varphi \leq 2\pi$$

But there are other conventions also commonly used:



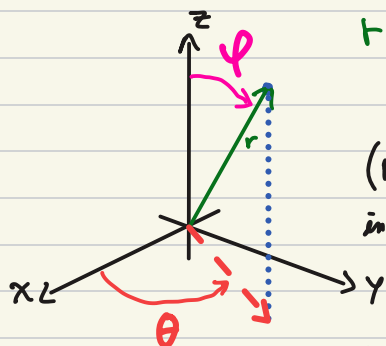
r - radius, θ - polar angle, φ - azimuthal angle

$$(r, \theta, \varphi) \leftrightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} r \sin \theta \cos \varphi \\ r \sin \theta \sin \varphi \\ r \cos \theta \end{pmatrix}$$

in polar in cartesian

$$0 \leq r, \quad 0 \leq \theta \leq \pi, \quad 0 \leq \varphi \leq 2\pi$$

The book's convention (or mathematic convention):



r - radius, φ - polar angle, θ - azimuthal angle

$$(r, \varphi, \theta) \leftrightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} r \sin \varphi \cos \theta \\ r \sin \varphi \sin \theta \\ r \cos \varphi \end{pmatrix}$$

in polar in cartesian

$$0 \leq r, \quad 0 \leq \varphi \leq \pi, \quad 0 \leq \theta \leq 2\pi$$

may also use ρ for radius

Change of coordinates:

- An "Base" system of coordinates in the Cartesian $(x, y, z, \dots)$
with:- length of an interval $[a, b]$ being $b-a = \int_a^b 1 dx$
- area of a simple region R in $\mathbb{R}^2$ is $\iint_R 1 dA$
 $dA = dx dy, dy dx$
- volume of a simple region W in $\mathbb{R}^3$ is $\iiint_W 1 dV$
 $dV = dx dy dz, dy dx dz, \dots$

def: A change of coordinate is a map $T: D^* \xrightarrow{\subseteq \mathbb{R}^n} D \subseteq \mathbb{R}^n$
which is C^1 (all partial derivatives continuous) and
one-to-one and surjective.

remark: For integration we can have maps that are not one-to-one
but the "size" (length, area, volume, etc.) of where the
map is not one-to-one is 0.

Types of coordinate transform:

- Translation: iff we have a region D^* in $\mathbb{R}^n$
we can apply a translation
 $T: D^* \rightarrow D \quad T(\vec{u}) = \vec{u} + \vec{b}$
where $\vec{b} \in \mathbb{R}^n$ is constant

This doesn't affect the volume of a region

e.g. : $[1, 3] \times [2, 5] \subseteq \mathbb{R}^2 \quad T(\vec{u}) = \vec{u} + (-1, -2)$

$D^* = [0, 2] \times [0, 3]$
 $\text{area}(D^*) = 6$

$D = T(D^*) = [0, 2] \times [0, 3]$
 $\text{area}(D) = 6$

• Linear Transformation: The transformation $T: D^* \rightarrow D$ $\subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$

is of the form $T(\vec{u}) = A\vec{u}$ where

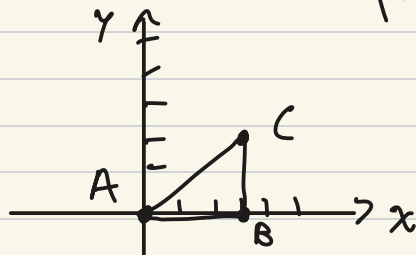
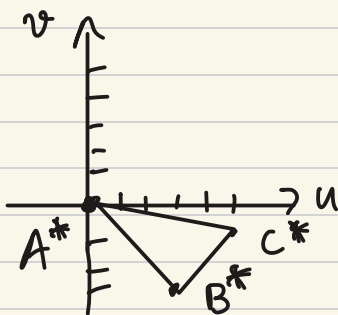
$$A \in M_{n \times n}(\mathbb{R}) \quad \text{and} \quad \det(A) \neq 0$$

$\uparrow$ constant $\downarrow$

implies $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$
is 1-to-1 and surjective

e.g. Let D^* be the triangle with vertices
 $A^*(0,0)$, $B^*(3,-3)$, $C^*(5,1)$

$$\text{and } T(u,v) = \left(\frac{u}{2} - \frac{v}{2}, \frac{u}{2} + \frac{v}{2} \right) \sim \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$$



$$\begin{aligned} A &= T(0,0) = 0 \\ B &= T(3,-3) = (3,0) \\ C &= T(5,-1) = (3,2) \end{aligned}$$

Proposition: If $T: D^* \rightarrow D$ is a linear change of coordinates, then:

- 1) Straight lines in D^* go to straight lines in D
- 2) A region $W^* \subseteq D^*$ with volume $\text{vol}(W^*)$ goes to a region $W = T(W^*)$ with volume $|\det(A)| \text{vol}(W^*)$ where A is the matrix of the linear transform

* This is an affine transformation (i.e. composition of translation and linear transformation)

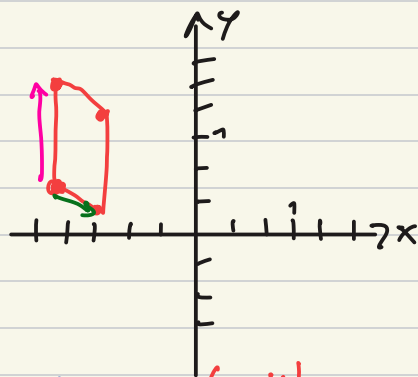
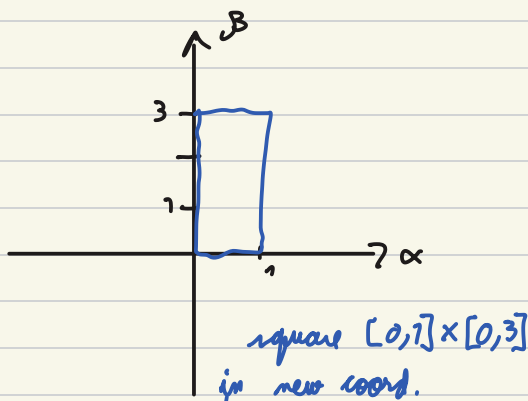
in 2-D: $\alpha = 2x + 3$, $\beta = x + 3y$

orig. coord. = (x, y) new coord. = (α, β)

$$T^{-1}(x, y) = (\alpha, \beta) = (2x + 3, x + 3y)$$

$$T(\alpha, \beta) = (x, y) = \left(\frac{\alpha - 3}{2}, \left(\beta - \frac{\alpha - 3}{2} \right) / 3 \right)$$

x



(α, β)

$(0, 0)$

$(1, 0)$

$(1, 3)$

$(0, 3)$

(x, y)

$\left(-\frac{3}{2}, \frac{1}{2} \right)$

$\left(-1, \frac{1}{3} \right)$

$\left(-1, \frac{4}{3} \right)$

$\left(-\frac{3}{2}, \frac{3}{2} \right)$

Area in (α, β) is 3

Area in (x, y) is $\det \begin{pmatrix} -1 + \frac{3}{2} & \frac{1}{3} - \frac{1}{2} \\ -\frac{3}{2} - \frac{3}{2} & \frac{3}{2} - \frac{1}{2} \end{pmatrix}$

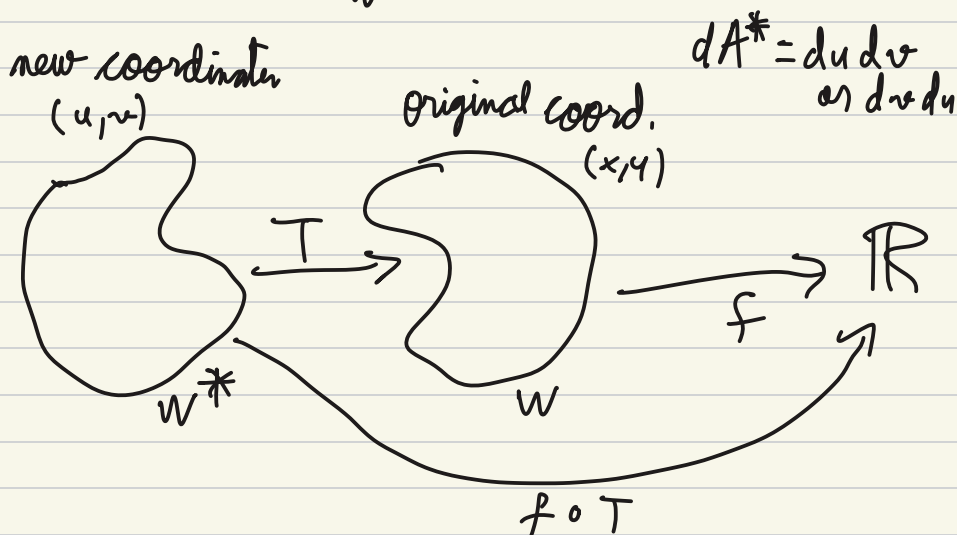
$= \det \begin{pmatrix} \frac{1}{2} & -\frac{1}{6} \\ 0 & 1 \end{pmatrix} = \frac{1}{2}$

$$DT = \begin{pmatrix} \frac{\partial x}{\partial \alpha} & \frac{\partial x}{\partial \beta} \\ \frac{\partial y}{\partial \alpha} & \frac{\partial y}{\partial \beta} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 0 \\ -\frac{1}{6} & \frac{1}{3} \end{pmatrix} = \frac{1}{6}$$

General idea of change of variable inside an integral

Original integral $\iint_W f(x, y) dA$

want new integral $\iint_{W^*} g(u, v) dA^*$



but we need to account for how T changes area

T theorem (change of variable in $\mathbb{R}^2$):

If $T: W^* \rightarrow W$ is a C^1 function, surjective and either one-to-one or region where this map is not one-to-one has zero area, then

if $f: W \rightarrow \mathbb{R}$ is integrable, then

$$\iint_W f(x, y) dx dy = \iint_{W^*} f \circ T |\det(DT)| du dv.$$

$$\begin{aligned}
 \text{e.g. } \int_0^1 \int_0^2 (x^2 + 2xy + y^2) dx dy &= \int_0^1 \left[\frac{x^3}{3} + x^2 y + y^2 x \right]_{x=0}^{x=2} dy \\
 &= \int_0^1 \left(\frac{8}{3} + 4y + 2y^2 \right) dy \\
 &= \left(\frac{8}{3} y + 2y^2 + \frac{2}{3} y^3 \right) \Big|_{y=0}^{y=1} = \frac{8}{3} + 2 + \frac{2}{3} \\
 &= \frac{10}{3} + 2 = \frac{16}{3}
 \end{aligned}$$

Now let's try the change of coord.
 $u = x+y$ and $v = x-2y$

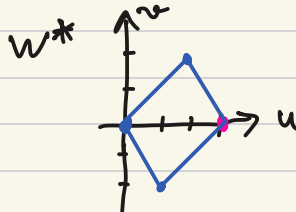
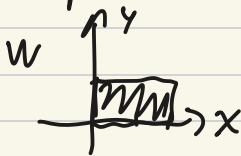
$$\begin{aligned}
 \text{new coord. } (u, v) &\xrightarrow{T} \text{orig. coord. } (x, y) \\
 u - v &= 3y & 2u + v &= 3x
 \end{aligned}$$

$$T(u, v) = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{2u+v}{3} \\ \frac{u-v}{3} \end{pmatrix} \quad \text{correction ahead}$$

$$DT = \frac{\partial(x, y)}{\partial(u, v)} = \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix} = \begin{pmatrix} 2/3 & 1/3 \\ 1/3 & -1/3 \end{pmatrix}$$

other notation for the jacobian matrix

$$\begin{aligned}
 \text{jacobian} = \det(DT) &= \left| \frac{\partial(x, y)}{\partial(u, v)} \right| = \frac{2}{3} \cdot \frac{-1}{3} - \frac{1}{3} \cdot \frac{1}{3} \\
 &= \frac{-3}{9} = -\frac{1}{3}
 \end{aligned}$$



x, y	u, v
$(0, 0)$	$(0, 0)$
$(2, 0)$	$(2, 2)$
$(2, 1)$	$(3, 0)$
$(0, 1)$	$(1, -2)$

$$\iint_W (x^2 + 2xy + y^2) dx dy = \iint_{W^*} u^2 \left(\frac{-1}{3} \right) du dv$$

The jacobian matrix for a coordinate transformation

$$T(u,v) = (x(u,v), y(u,v))$$

in the matrix of partial derivatives

$$DT = \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{bmatrix}.$$

The jacobian is the determinant of DT and we sometimes use the notation $\frac{\partial(x,y)}{\partial(u,v)} = \det(DT)$

and the change of variables $T: W^* \rightarrow W$

transform the integral of f on W by

$$\iint_W f(x,y) \, dx \, dy = \iint_{W^*} f \circ T(u,v) \, |\det(DT)| \, du \, dv$$

on $dx \, dy$ on $du \, dv$